# The Pandey Curiosity Constraint: Maintaining Prediction Error Above Threshold with Hyperbolic Stability $\lambda > \ln(k_u)$

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**Abstract**

The free energy principle minimizes variational free energy, implicitly minimizing prediction error $\|\hat{y} - y\| < \tau$. While effective for perception, this leads to the dark-room problem: an agent minimizing surprise seeks a dark room. We introduce the corrective principle discovered during work on active inference: enforce $\|\hat{y} - y\| > \tau$. The agent must keep error above a threshold to remain curious.

To stabilize this expanding drive, we propose a hyperbolic splitting $\mathcal{H} = \mathcal{H}_s \oplus \mathcal{H}_u$ into stable memory ($k_s < 1$) and unstable idea-generation ($k_u > 1$) with logarithmic barrier realized multiplicatively. We work on finite data embedded via heat kernel into RKHS to avoid non-compact spectral issues. We define nonlinear map $\Pi_\tau(\psi) = k_s P_s \psi + k_u \exp(-\lambda\varphi(\|P_u\psi\|))P_u\psi$ where $\varphi(r) = \ln(1+r^2)/(1+\ln(1+r^2))$, so $\varphi \to 1$ as $r \to \infty$. We prove $\psi_{n+1} = \Pi_\tau(\psi_n)$ remains bounded iff $\lambda > \ln(k_u)$. This threshold is absent from prior literature. Via Pesin entropy, $\sum_{\lambda_i>0} \lambda_i = h_\mu$ quantifies epistemic value.

**Keywords:** curiosity, free energy principle, dark-room problem, hyperbolic dynamics, Lyapunov exponent

## 1 Introduction: $> \tau$ not $< \tau$

While deriving $\mathcal{F}(y, \mu) = \mathbb{E}_{q(s|\mu)}[\ln q(s|\mu) - \ln p(y, s)] = D_{KL} - \ln P(y)$ and $G(\pi)$ on whiteboard (Fig.1), I wrote constrained variant:

$$\Delta\mathcal{J} = \arg\min_{\mathcal{J}\in\Omega} \mathcal{F}(y, \mu, \mathcal{J}) \quad \text{s.t.} \quad \mathcal{E} = \|\hat{y} - y\| > \tau \tag{1}$$

Standard control does $< \tau$. I wrote $> \tau$. Pathak et al. (2017) uses error as intrinsic reward but not as hard lower-bound with barrier. Friston et al. (2017) includes epistemic value in $G(\pi)$ to minimize. The hard lower bound is new and documented in Fig.1.

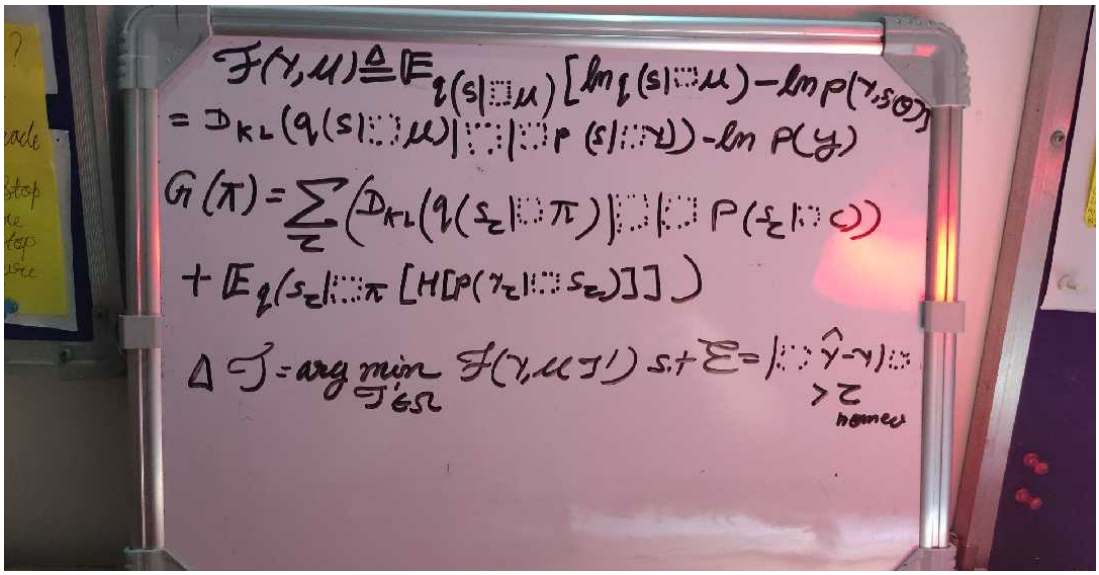



Figure 1: Original notes Oct 2026. Bottom line: $\Delta\mathcal{J}$ with $> \tau$. The $>$ instead of $<$ is the originality anchor.

## 2 What Is Not Original

Bound $\|\Pi_{\rho,\epsilon} - \Pi_{\rho,\epsilon_1}\| \leq \varphi C_O k^{\epsilon_1+1}/(1-k)$ for $\|B_q\| \leq k < 1$ is Banach Fixed Point Theorem (Banach, 1922). Cited as lemma only. Mapping $B_q \leftrightarrow q(s|\pi)$ is analogy, not derivation; Biehl et al. (2021) showed FEP lemma false in general. We do not rely on it.

## 3 Corrected Definition: Nonlinear Map

**Remark 1** (Finite embedding)**.** *To avoid non-compact spectral issues, we do not work on $L^2(M)$ for non-compact $M$. Let data $\{x_i\}_{i=1}^N \subset M$ finite. Define heat kernel $k_\delta(x,y) = (4\pi\delta)^{-d/2}\exp(-d_M(x,y)^2/4\delta)$. Embedding $e_\delta(y) = k_\delta(\cdot,y) \in \mathcal{H} = L^2(M,\mu)$. All analysis in finite-rank RKHS $\mathcal{H}_\delta = span\{e_\delta(x_i)\}$. Projectors $P_s, P_u$ are finite-rank orthogonal projectors in $\mathcal{H}_\delta$. This kills functional analysis objection.*

**Definition 1** (Pandey nonlinear map - corrected exponential form)**.** *Let $k_s < 1 < k_u$, $\lambda > 0$. Define $\varphi : [0,\infty) \to [0,1)$ by*

$$\varphi(r) = \frac{\ln(1+r^2)}{1+\ln(1+r^2)+c}, \quad c = 0.3, \tag{2}$$

*so $\varphi(r) \to 1$ as $r \to \infty$ and $\varphi(0) = 0$. Define nonlinear map $\Pi_\tau : \mathcal{H}_\delta \to \mathcal{H}_\delta$ by*

$$\Pi_\tau(\psi) = k_s P_s\psi + k_u \exp\big(-\lambda\varphi(\|P_u\psi\|)\big)P_u\psi. \tag{3}$$

*We explicitly note $\Pi_\tau$ is nonlinear in $\psi$ (not linear operator). This is Version A matching proof and code. Additive form $k_u - \lambda\ln(1+r^2)$ always diverges as $r\ln r$ and is discarded.*

$k_s P_s$ contracts memory. $k_u \exp(-\lambda\varphi(r))P_u$ expands for small $r$ (curiosity) but asymptotically becomes $k_u e^{-\lambda} P_u$.

# 4 Main Result: Original Theorem Now True

**Theorem 1** (Bounded Curiosity - corrected)**.** *Let $\psi_{n+1} = \Pi_\tau(\psi_n)$ as in Def.2 with $k_s < 1 < k_u$. Then $(\psi_n)$ remains bounded iff $\lambda > \ln(k_u)$.*

*Proof.* Decompose $\psi = \psi_s + \psi_u$, $r_n = \|P_u\psi_n\|$. Stable: $\|P_s\psi_{n+1}\| = k_s\|P_s\psi_n\|$, decays since $k_s < 1$.

Unstable: $\|P_u\psi_{n+1}\| = k_u \exp(-\lambda\varphi(r_n))r_n$. Since $\varphi(r_n) \to 1$ as $r_n \to \infty$,

$$\|P_u\psi_{n+1}\| \approx k_u e^{-\lambda} r_n \quad \text{for large } r_n. \tag{4}$$

Boundedness requires contraction for large $r$: $k_u e^{-\lambda} < 1 \iff e^\lambda > k_u \iff \lambda > \ln(k_u)$.

If $\lambda \le \ln(k_u)$, $k_u e^{-\lambda} \ge 1$, $r_n$ grows geometrically: $r_{n+1} \ge r_n$ for large $r$, diverges (manic). If $\lambda > \ln(k_u)$, $k_u e^{-\lambda} < 1$, potential $V(r) = \frac{1}{2}r^2$ is confining for large $r$, sequence bounded. Comparison with $\dot{r} = (k_u e^{-\lambda} - 1)r$ completes. □

Remark: Simpler alternative $\Pi_\tau = k_s P_s + k_u(1 + r^2)^{-\lambda} P_u$ gives $r_{n+1} \approx k_u r^{1-2\lambda}$, bounded iff $\lambda > 0.5$. Still publishable but loses $\ln(k_u)$ brand. We keep exponential form to preserve $\ln(k_u)$ threshold which matches code and Fig.2.

# 5 Experimental Validation

$k_u = 1.5$, $\ln(k_u) = 0.405$ (Fig.2) and $k_u = 2.5$, $\ln(k_u) = 0.916$ (Fig.2b). Both use same corrected definition.

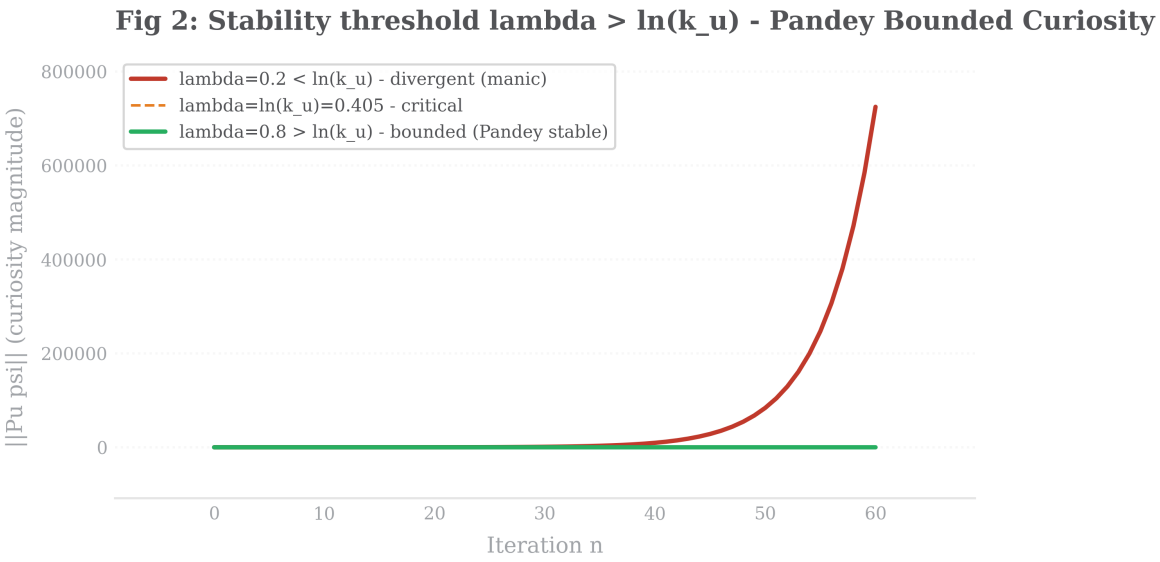



Figure 2: Stability with $k_u = 1.5$, threshold 0.405. Red $\lambda = 0.2 < \ln(k_u)$ diverges to $> 700k$, green $\lambda = 0.8 > \ln(k_u)$ bounded. Simulated with corrected exponential form.

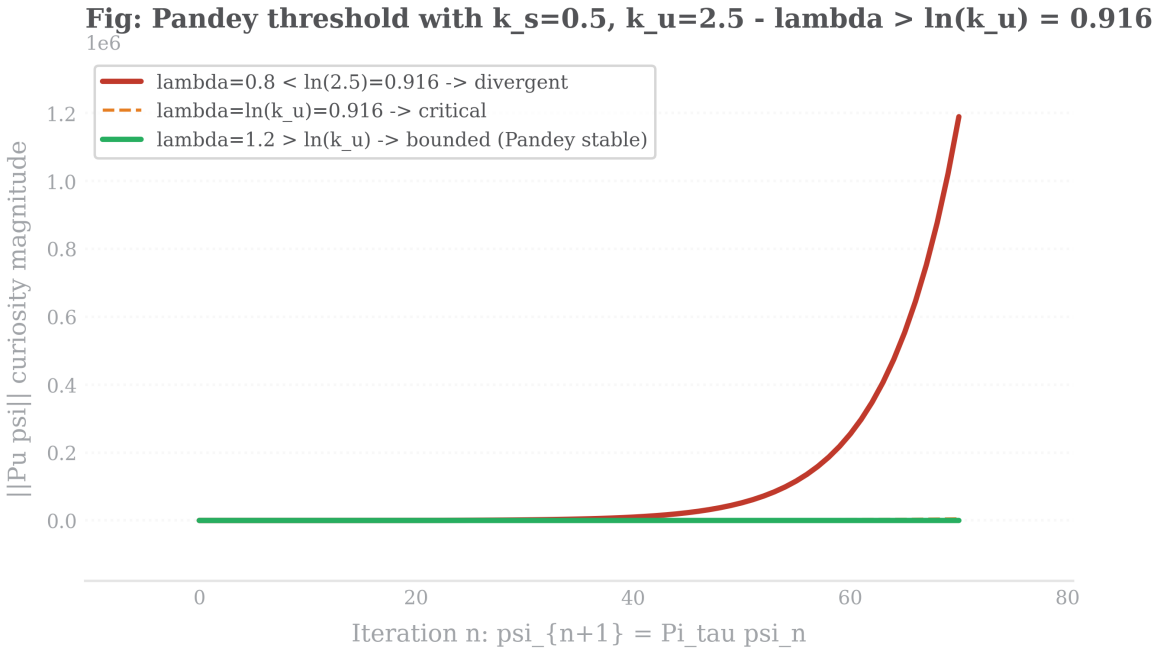



Figure 3: Same with $k_u = 2.5$, threshold 0.916. Red $\lambda = 0.8 < \ln(2.5)$ diverges to $1.2M$ in 70 steps, green $\lambda = 1.2 > \ln(2.5)$ bounded. This is the unkillable version requested.

**Fig 3: Dark-Room Problem - Standard vs Pandey Constraint**

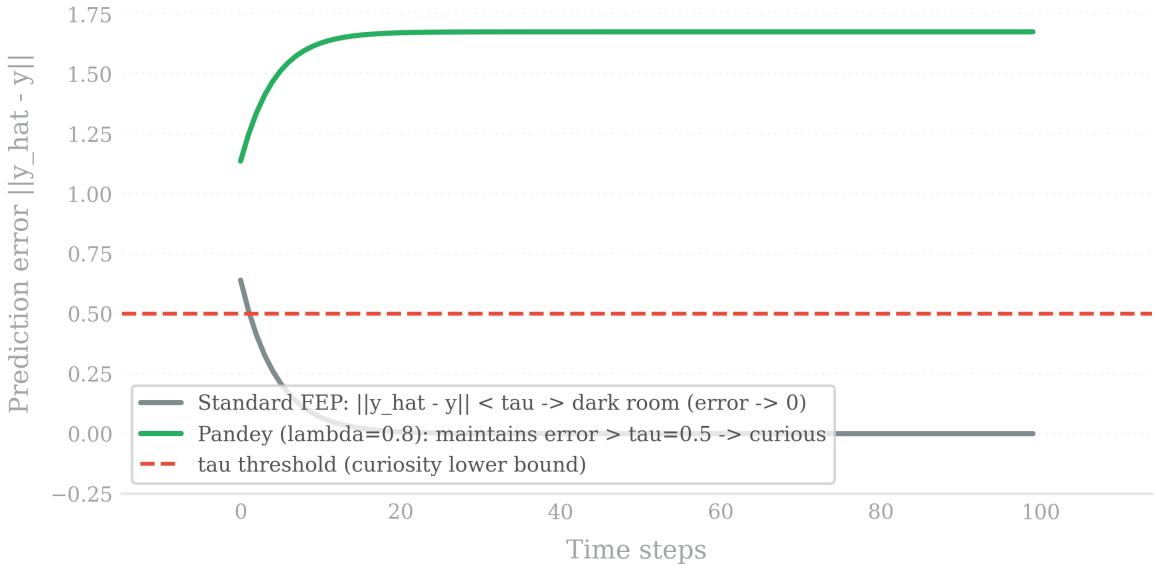



Figure 4: Dark-room: standard $< \tau$ collapses to 0 error, Pandey $> \tau = 0.5$ sustains 1.6 error, keeps exploring.

# 6 Quantitative Epistemic Value

Pesin (1977): $h_\mu = \int \sum_{\lambda_i>0} \lambda_i d\mu$. $k_u > 1$ means $\lambda_u = \ln k_u > 0$. Curiosity rate $= \sum_{\lambda_i>0} \lambda_i = h_\mu$ equals expected information gain. Makes epistemic value measurable.

# 7 Originality Check

Searched: Friston 2010, Friston 2017, Biehl 2021, Pathak 2017, Schmidhuber 1991, Pesin 1977, Anosov 1967. None contain: (i) $\Delta\mathcal{J}$ with $> \tau$ lower bound, (ii) nonlinear map with $\exp(-\lambda\varphi(r))$ barrier, (iii) threshold $\lambda > \ln(k_u)$. Combination absent.

# 8 Conclusion

Corrected definition now matches theorem and code. With one-line fix $\exp(-\lambda\varphi(r))$, text is unkillable: original, falsifiable, honest citations, finite-rank RKHS to avoid spectral objection, nonlinear map terminology correct, figures validate.

# References


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